A Two-Sided Schur Certificate for Packet-to-Riesz Lifting

Directed Feedback Beyond the Symmetric Neumann Gate

Prime Dynamics Theory Program

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Abstract

RH-161 certifies a packet-to-Riesz lift from the symmetric condition $M\|E\| < 1$. For an orthogonal packet decomposition the off-diagonal operator has norm $\max(\|B\|, \|C\|)$, so this test is governed by the larger directed coupling even when the return coupling is tiny. We prove a strictly sharper Schur-feedback theorem.

On a separating contour let a, d bound the packet and complement resolvents, and put $b = \|B\|$, $c = \|C\|$. If

$$\kappa = adbc < 1,$$

the contour remains in the resolvent of every off-diagonal homotopy, hence the enclosed Riesz rank equals the packet rank. We give an explicit scalar 2×2 bound for the full projector error. The criterion certifies families with $b \to \infty$, $c \to 0$, and bc bounded; triangular systems have zero feedback regardless of the surviving coupling. A 512-matrix audit has no bound failures. This is certificate technology: no physical all-level contour or coupling estimate is asserted.

1 Directed block data

Let P be a finite-rank orthogonal projection, $Q = I - P$, and write

$$A = \begin{pmatrix} A_P & B \\ C & A_Q \end{pmatrix}, \qquad A_t = \begin{pmatrix} A_P & tB \\ tC & A_Q \end{pmatrix}, \quad 0 \le t \le 1.$$

Let Γ be a positively oriented rectifiable Jordan curve enclosing $\sigma(A_P)$ and excluding $\sigma(A_Q)$. Set

$$X(z) = (z - A_P)^{-1}, \quad Y(z) = (z - A_Q)^{-1},$$

and assume

$$\sup_\Gamma \|X\| \le a, \qquad \sup_\Gamma \|Y\| \le d, \qquad \|B\| \le b, \quad \|C\| \le c.$$

The letters B and C are ordered: C sends the packet outward and B returns the complement to the packet. RH-162 shows how primal and adjoint ambient defects can supply these two numbers [Program, 2026b].

2 Schur-feedback rank preservation

Theorem 1 (Two-sided Schur packet lift). *If $\kappa = adbc < 1$, then $\Gamma \subset \rho(A_t)$ for every $t \in [0,1]$. The Riesz projection*

$$\Pi = \frac{1}{2\pi i}\int_{\Gamma}(z-A)^{-1}\,dz$$

has $\operatorname{rank}\Pi = \operatorname{rank}P$.

Proof. For $z \in \Gamma$, eliminate the complementary block. The packet Schur complement is

$$S_t(z) = (z-A_P) - t^2BY(z)C = (z-A_P)\big(I - t^2X(z)BY(z)C\big).$$

The second factor is invertible because $\|t^2XBYC\| \le t^2\kappa < 1$. Since $z - A_Q$ is invertible, the block factorization makes $z - A_t$ invertible. The Riesz projections vary continuously with t and have constant finite rank. At $t = 0$ the projection is P, proving the claim [Kato, 1995]. □

The condition is invariant under reciprocal scalar rescaling of the two blocks. It records the closed feedback loop $P \to Q \to P$, rather than the largest one-way excursion.

3 An explicit projector bound

For a scalar matrix $H = (h_{ij})_{i,j=1}^2$, write $\|H\|_2$ for its Euclidean operator norm.

Theorem 2 (Block resolvent and projector envelope). *Under Theorem 1, for every $z \in \Gamma$,*

$$\|(z-A)^{-1} - (z-A_0)^{-1}\| \le \frac{1}{1-\kappa}\left\|\begin{pmatrix} a\kappa & abd \\ acd & d\kappa\end{pmatrix}\right\|_2.$$

Consequently

$$\|\Pi - P\| \le \frac{|\Gamma|}{2\pi(1-\kappa)}\left\|\begin{pmatrix} a\kappa & abd \\ acd & d\kappa\end{pmatrix}\right\|_2 =: \delta_S.$$

If $\delta_S < 1$, the Riesz range is a graph over the packet with slope at most $\delta_S/(1-\delta_S)$.

Proof. The Schur inverse formula and the Neumann series give

$$\|R_{11} - X\| \le a\kappa/(1-\kappa), \qquad \|R_{12}\| \le abd/(1-\kappa),$$
$$\|R_{21}\| \le acd/(1-\kappa), \qquad \|R_{22} - Y\| \le d\kappa/(1-\kappa).$$

The norm of an operator block matrix is bounded by the norm of the scalar matrix of block bounds. Contour integration proves the projector estimate. The graph conclusion is the standard close-projection argument used in RH-161 [Program, 2026a]. □

4 Strict improvement and exact limiting cases

The symmetric RH-161 condition, with the exact off-diagonal norm, is

$$\max(a, d) \max(b, c) < 1.$$

Take $a = d = 1$, $b = L$, and $c = \theta/L$ with $0 < \theta < 1$. Then the Schur product is $\kappa = \theta$ for every L, while the symmetric product equals $\max(L, \theta/L)$ and diverges. Thus the improvement is unbounded.

If $C = 0$, the packet is invariant and the block matrix is upper triangular. The feedback product is zero, so Theorem 1 preserves the packet spectrum for arbitrary B. The Riesz projection may be highly oblique, and the projector bound correctly still contains abd. Rank preservation and graph conditioning are different conclusions.

5 Pointwise sharpness of the feedback threshold

The constant one in the Schur test cannot be improved from norm data alone. Consider scalar blocks at a fixed contour point z_0:

$$z_0 - A = \begin{pmatrix} x^{-1} & -b \\ -c & y^{-1} \end{pmatrix}, \qquad |x| = a, \quad |y| = d.$$

Proposition 3 (Threshold contact witness)**.** *For phases chosen so that $xybc = 1$, the point z_0 belongs to $\sigma(A)$. Hence the non-strict condition $adbc \leq 1$ cannot certify a resolvent contour uniformly over the displayed norm class.*

Proof. The determinant of $z_0 - A$ equals $x^{-1}y^{-1} - bc$. It vanishes when $xybc = 1$. Phases of the scalar couplings can always realize the equality whenever $adbc = 1$. □

This witness is local: a physical certificate needs a strict margin at every point of the contour. An average value of $adbc$, or a favorable value on a dense but uncertified mesh, does not exclude a single threshold contact. It also explains why the finite-mesh denominator developed later must be strictly below one.

6 Audit and boundary

The companion calculation samples 512 finite block systems. It synthesizes block resolvents X, Y, forms the exact full inverse, and compares its difference from $X \oplus Y$ with the scalar envelope of Theorem 2. Every accepted sample satisfies the bound. The audit is a regression test, not validated prime-dynamics data.

RH-163 proves that the first rank gate inside physical interface R should be tested with $adbc$, not a symmetric maximum. It does not provide the ambient realization, the contour, any of the four physical bounds, a uniform scale margin, the complementary Schatten limit, Gate A, or any Hilbert–Polya or zeta conclusion.

References

Tosio Kato. *Perturbation Theory for Linear Operators*. Springer, 1995.

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Prime Dynamics Theory Program. The ambient-realization gate between reset packets and riesz clouds, 2026b. RH-162 manuscript.